

miniprojectml166

```
In [1]:
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.svm import SVC
from sklearn.metrics import accuracy_score, classification_report, confusion_matrix
```

```
In [2]:
data = pd.DataFrame({
    'Resume': [
        'Python machine learning pandas numpy data analysis',
        'Python developer Django Flask API programming',
        'Machine learning deep learning Python TensorFlow',

        'HTML CSS JavaScript React frontend web development',
        'Web developer HTML CSS JavaScript Bootstrap',
        'React JavaScript Node.js frontend developer',

        'Java Spring Boot Hibernate SQL backend development',
        'Java developer Spring framework JDBC SQL',
        'Core Java OOP Spring Boot developer',

        'Python Django Flask programming developer',
        'Python automation scripting Django developer',
        'Python NumPy Pandas machine learning'
    ],
    'Category': [
        'Data Science',
        'Python Developer',
        'Data Science',

        'Web Development',
        'Web Development',
        'Web Development',

        'Java Developer',
        'Java Developer',
        'Java Developer',

        'Python Developer',
        'Python Developer',
        'Data Science'
    ]
})

data
```

	Resume	Category
0	Python machine learning pandas numpy data anal...	Data Science
1	Python developer Django Flask API programming	Python Developer
2	Machine learning deep learning Python TensorFlow	Data Science
3	HTML CSS JavaScript React frontend web develop...	Web Development
4	Web developer HTML CSS JavaScript Bootstrap	Web Development
5	React JavaScript Node.js frontend developer	Web Development
6	Java Spring Boot Hibernate SQL backend develop...	Java Developer
7	Java developer Spring framework JDBC SQL	Java Developer
8	Core Java OOP Spring Boot developer	Java Developer
9	Python Django Flask programming developer	Python Developer
10	Python automation scripting Django developer	Python Developer
11	Python NumPy Pandas machine learning	Data Science

```
In [3]:
data.head()
```

	Resume	Category
0	Python machine learning pandas numpy data anal...	Data Science
1	Python developer Django Flask API programming	Python Developer
2	Machine learning deep learning Python TensorFlow	Data Science
3	HTML CSS JavaScript React frontend web develop...	Web Development
4	Web developer HTML CSS JavaScript Bootstrap	Web Development

```
In [4]:
data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 12 entries, 0 to 11
Data columns (total 2 columns):
#   Column      Non-Null Count  Dtype
---  ---
0    Resume      12 non-null     object
1    Category     12 non-null     object
dtypes: object(2)
memory usage: 320.0+ bytes
```

In [5]:

```
data.isnull().sum()
```

```
Resume      0
Category     0
dtype: int64
```

In [6]:

```
print("Number of Rows:", data.shape[0])
print("Number of Columns:", data.shape[1])
```

```
Number of Rows: 12
Number of Columns: 2
```

In [7]:

```
print(data['Category'].value_counts())
```

```
Category
Data Science      3
Python Developer  3
Web Development   3
Java Developer     3
Name: count, dtype: int64
```

In [8]:

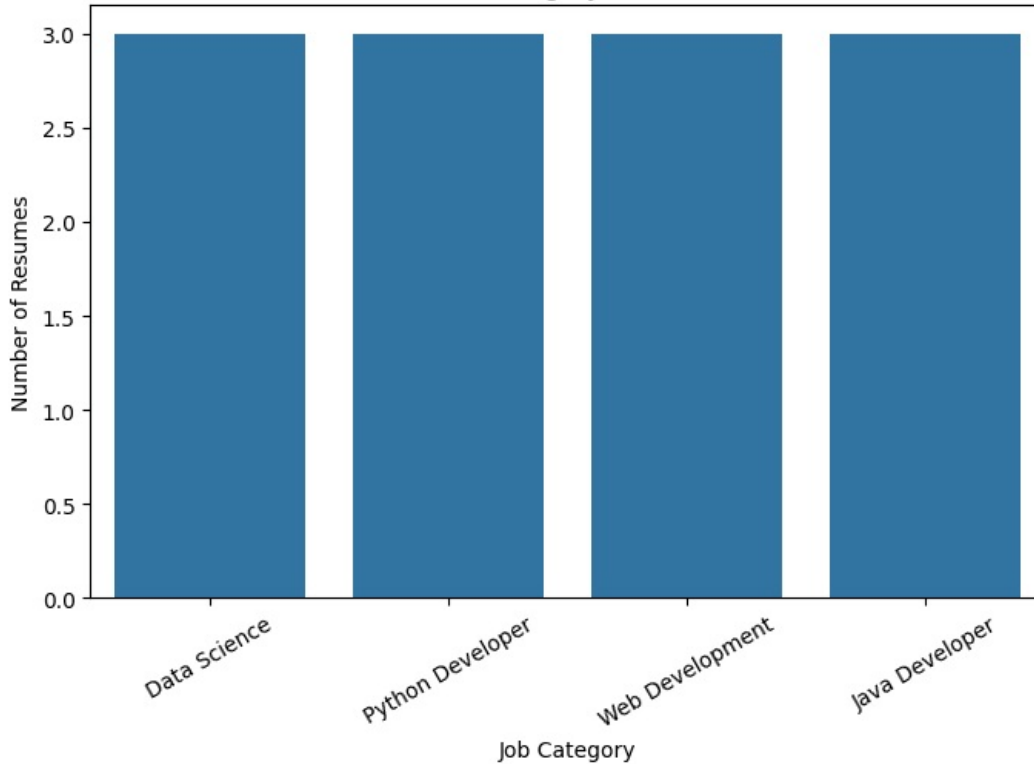
```
plt.figure(figsize=(8,5))

sns.countplot(x='Category', data=data)

plt.xticks(rotation=30)
plt.title("Resume Category Distribution")
plt.xlabel("Job Category")
plt.ylabel("Number of Resumes")

plt.show()
```

Resume Category Distribution



In [9]:

```
X = data['Resume']
Y = data['Category']

print("Input:")
print(X.head())

print("\nOutput:")
print(Y.head())
```

```
Input:
0    Python machine learning pandas numpy data anal...
1    Python developer Django Flask API programming
2    Machine learning deep learning Python TensorFlow
3    HTML CSS JavaScript React frontend web develop...
4    Web developer HTML CSS JavaScript Bootstrap
Name: Resume, dtype: object

Output:
0    Data Science
1    Python Developer
2    Data Science
3    Web Development
4    Web Development
Name: Category, dtype: object
```

In [10]:

```
vectorizer = TfidfVectorizer()

X = vectorizer.fit_transform(data['Resume'])

print("TF-IDF conversion completed")
print("Feature Shape:", X.shape)
```

```
TF-IDF conversion completed
Feature Shape: (12, 36)
```

In [11]:

```
X_train, X_test, Y_train, Y_test = train_test_split(
    X,
    Y,
    test_size=0.25,
    random_state=42
)

print("Training Data:", X_train.shape)
print("Testing Data:", X_test.shape)
```

```
Training Data: (9, 36)
Testing Data: (3, 36)
```

In [12]:

```
model = SVC(
    kernel='linear',
    random_state=42
)

model.fit(X_train, Y_train)

print("Model Training Completed")
```

```
Model Training Completed
```

In [13]:

```
Y_pred = model.predict(X_test)

print("Prediction Completed")
```

```
Prediction Completed
```

In [14]:

```
accuracy = accuracy_score(Y_test, Y_pred)

print("Model Accuracy:", accuracy)
print("Accuracy Percentage:", accuracy * 100, "%")
```

```
Model Accuracy: 0.6666666666666666
Accuracy Percentage: 66.66666666666666 %
```

In [15]:

```
print(classification_report(Y_test, Y_pred, zero_division=0))
```

	precision	recall	f1-score	support
Data Science	1.00	1.00	1.00	1
Python Developer	1.00	0.50	0.67	2
Web Development	0.00	0.00	0.00	0
accuracy			0.67	3
macro avg	0.67	0.50	0.56	3
weighted avg	1.00	0.67	0.78	3

In [16]:

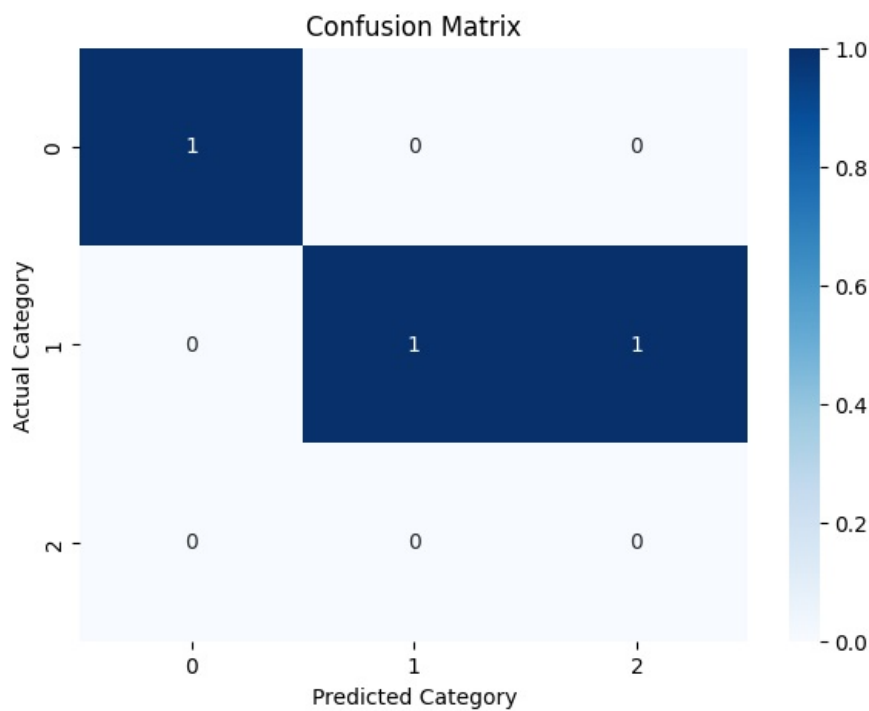
```
cm = confusion_matrix(Y_test, Y_pred)

plt.figure(figsize=(7,5))

sns.heatmap(
    cm,
    annot=True,
    fmt='d',
    cmap='Blues'
)

plt.title("Confusion Matrix")
plt.xlabel("Predicted Category")
plt.ylabel("Actual Category")

plt.show()
```



In [17]:

```
new_resume = """
Python developer with experience in Django, Flask,
REST API, Python programming and SQL
"""

resume_vector = vectorizer.transform([new_resume])

prediction = model.predict(resume_vector)

print("Predicted Resume Category:", prediction[0])
```

```
Predicted Resume Category: Python Developer
```

In [18]:

```
def screen_resume(resume):

    resume_vector = vectorizer.transform([resume])

    prediction = model.predict(resume_vector)

    print("Resume Screening Result")
    print("-----")
    print("Predicted Category:", prediction[0])
```

In [20]:

```
resume = input("Enter Resume Skills / Description: ")

screen_resume(resume)
```

```
Enter Resume Skills / Description: I know HTML CSS JavaScript React and frontend development
```

```
Resume Screening Result
-----
Predicted Category: Web Development
```

In [21]:

```
resume = input("Enter Resume Skills / Description: ")

screen_resume(resume)
```

Enter Resume Skills / Description: Machine learning Python pandas numpy TensorFlow

Resume Screening Result

Predicted Category: Data Science